A single correlation clustering that approximates every disagreement l_p norm simultaneously

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Editorial note. Only the simultaneous all-norms main theorem counts. Complete original mathematical body and references retained, inputs mechanically expanded. Original cc.bib is rendered via installed natbib/BibTeX; original biblatex citation metadata remains preserved in native files. Abstract wrapper uses an explicit environment without changing its text.

Allowed standard prerequisites (editorial).

- Standard algorithmic and mathematical background cited by the authors. All source-local constructions, proofs, appendices and references are retained; no new mathematical assumptions added.

Abstract

This paper considers correlation clustering on unweighted complete graphs. We give a combinatorial algorithm that returns a single clustering solution that is simultaneously $O(1)$-approximate for all ℓ_p-norms of the disagreement vector; in other words, a combinatorial $O(1)$-approximation of the all-norms objective for correlation clustering. This is the first proof that minimal sacrifice is needed in order to optimize different norms of the disagreement vector. In addition, our algorithm is the first combinatorial approximation algorithm for the ℓ_2-norm objective, and more generally the first combinatorial algorithm for the ℓ_p-norm objective when $1 < p < \infty$. It is also faster than all previous algorithms that minimize the ℓ_p-norm of the disagreement vector, with run-time $O(n^\omega)$, where $O(n^\omega)$ is the time for matrix multiplication on $n \times n$ matrices. When the maximum positive degree in the graph is at most Δ, this can be improved to a run-time of $O(n\Delta^2 \log n)$.

1 Introduction

Correlation clustering is one of the most prominent problems in clustering, as it cleanly models community detection problems [37, 35] and provides a way to decompose complex network structures [38, 31]. The input to the unweighted correlation clustering problem is a complete graph $G = (V, E)$, where $|V| = n$ and each edge $e \in E$ is labeled positive $(+)$ or negative $(-)$. If the edge (u, v) is positive, this indicates that u and v are similar, and analogously if the edge (u, v) is negative, this indicates that u and v are dissimilar. The output of the problem is a partition of the vertex set into parts $C_1, C_2, \ldots$, where each part represents a cluster.

The output should cluster similar vertices together and separate dissimilar vertices. Specifically, for a fixed clustering (i.e., partition of the vertices), a positive edge (u, v) is a disagreement with respect to the clustering if u and v are in different clusters and an agreement if u and v are in the same cluster. Similarly, a negative edge (u, v) is a disagreement with respect to the clustering if

u and v are in the same cluster and an agreement if u and v are in different clusters. The goal is to find a clustering that minimizes some objective that is a function of the disagreements.[1] For example, the most commonly studied objective minimizes the total number of disagreements.

As an easy example to illustrate the problem, consider a social network. Every pair of people has an edge between them, and the edge is positive if the two people have ever met before, and negative otherwise. The goal of correlation clustering translates to partitioning all the people into clusters so that people are in the same cluster as their friends/acquaintances and in different clusters than strangers. The difficulty in constructing a clustering is that the labels may not be consistent, making disagreements unavoidable. Consider in the social network what happens when there is one person with two friends who have never met each other (u, v, w with (u, v) and (u, w) positive but (v, w) negative). The choice of objective matters in determining the best clustering.

For a given clustering $\mathcal{C}$, let $y_{\mathcal{C}}(u)$ denote the number of edges incident to u that are disagreements with respect to $\mathcal{C}$ (we drop $\mathcal{C}$ and write y when it is clear from context). The most commonly considered objectives are $\|y_{\mathcal{C}}\|_p = \sqrt[p]{\sum_{u \in V} y_{\mathcal{C}}(u)^p}$ for $p \in \mathbb{R}_{\geqslant 1} \cup \{\infty\}$, the ℓ_p-norms of the disagreement vector y. Note that the optimal objective values may drastically vary for different norms too. (For instance, in the example in Appendix A of [34], $V = A \sqcup B$[2], where $|A| = |B| = n/2$, and all edges are positive except for a negative matching between A and B. The optimal ℓ_∞-norm objective value is 1 whereas the optimal for ℓ_1 is $\Theta(n)$.) When $p = 1$, this objective minimizes the total number of disagreements. Setting $p = \infty$ minimizes the maximum number of disagreements incident to any node, ensuring a type of worst-case fairness.[3] Balancing these two extremes—average welfare on one hand and fairness on the other—is the ℓ_2-norm, which minimizes the variance of the disagreements at each node.

Correlation clustering was proposed by Bansal, Blum, and Chawla [2004] with the objective of minimizing the ℓ_1-norm of the disagreement vector. The problem is NP-hard and several approximation algorithms have been proposed [7, 4, 16, 18]. Puleo and Milenkovic [2016] proposed studying ℓ_p-norms of the disagreement vector for $p > 1$, and they give a 48-approximation for any fixed p. Charikar, Gupta, and Schwartz [2017] introduced an improved 7-approximation, which Kalhan, Makarychev, and Zhou [2019] further improved to a 5-approximation. When $p > 1$, up until recently, the only strategies were LP or SDP rounding, and it has been of interest to develop fast combinatorial algorithms [36]. Davies, Moseley, and Newman [2023] introduced a combinatorial $O(1)$-approximation algorithm for $p = \infty$ (see also [24] for a different combinatorial algorithm), and leave open the question of discovering a combinatorial $O(1)$-approximation algorithm for $1 < p < \infty$.

In all prior work, solutions obtained for ℓ_p-norms are tailored to each norm (i.e., p is part of the input to the algorithm), and it was not well-understood what the trade-offs were between solutions that optimize different norms. Solutions naively optimizing one norm can be arbitrarily bad for other norms (see Figure 1). A natural question is whether this loss from using a solution to one objective for another is avoidable. More specifically:

For any graph input to unweighted, complete correlation clustering, does there exist a partition (clustering) that is simultaneously $O(1)$-approximate for all ℓ_p-norm objectives?

Phrased another way, does there exist a universal algorithm for ℓ_p-norm correlation clustering—one which is guaranteed to produce a solution that well-approximates many objectives at once?

[1] Note that the sizes and number of clusters are unspecified.

[2] $\sqcup$ denotes disjoint union.

[3] In the social network example, minimizing the ℓ_1-norm corresponds to finding a clustering that minimizes the total number of friends who are separated plus the total number of strangers who are in the same cluster. The ℓ_∞-norm corresponds to finding a clustering minimizing the number of friends any person is separated from plus the number of strangers in that person's same cluster.

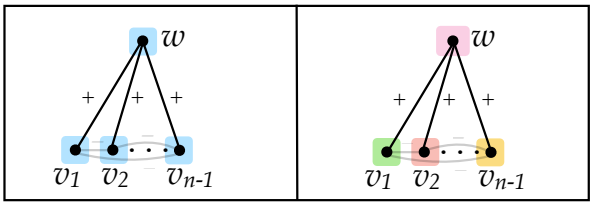


Figure 1: Two clusterings of the star graph, which has one node (w) with positive edges to all nodes, and the rest of the edges negative. Left: Clustering assigns all nodes to one (blue) cluster, and is (almost) optimal for the ℓ_∞-norm with cost $\Theta(n)$. Right: Clustering assigns all nodes to different clusters and is (almost) optimal for the ℓ_1-norm with cost $\Theta(n)$. The left solution is terrible for the ℓ_1-norm, as the negative clique has $\Theta(n^2)$ edges that are disagreements.

When the goal is to simultaneously optimize every ℓ_p-norm, this is known as the all-norms objective[4]. Universal algorithms and the all-norms objective are well-studied in combinatorial optimization problems, such as load balancing and set cover (see Section 1.2 for more discussion). In the context of correlation clustering, such an algorithm outputs a partition that has good global performance (i.e. ℓ_1-norm) and also has no individual node with too many adjacent disagreements (i.e. ℓ_∞-norm). Universal algorithms exist for some problems and are provably impossible for others. The question looms, what can be said about universal algorithms for correlation clustering?

As far as we are aware, there are no known results for the all-norms objective in other clustering problems. In fact, for the popular k-median and k-center problems, it is actually impossible to $O(1)$-approximate (or even $o(\sqrt{n})$-approximate) these two objectives simultaneously [5].

1.1 Results

This paper is focused on optimizing all ℓ_p-norms ($p \geqslant 1$) for correlation clustering at the same time. The main result of the paper answers the previous question positively: perhaps surprisingly, there is a single clustering that simultaneously $O(1)$-approximates the optimal for all ℓ_p-norms. Further, it can be found through an efficient combinatorial algorithm. This is also the first known combinatorial approximation algorithm for the ℓ_2-norm objective and more generally ℓ_p-norm objective for fixed $2 \leqslant p < \infty$.

In what follows, let $O(n^\omega)$ denote the run-time of $n \times n$ matrix multiplication.

Theorem 1. Let $G = (V, E)$ be an instance of unweighted, complete correlation clustering on $|V| = n$ nodes. There exists a combinatorial algorithm returning a single clustering that is simultaneously an $O(1)$-approximation[5] for all ℓ_p-norm objectives, for all $p \in \mathbb{R}_{\geqslant 1} \cup \{\infty\}$, and its run-time is $O(n^\omega)$.

The algorithm gives the fastest run-time of any $O(1)$-approximation algorithm for the ℓ_p-norm objective when $p \in \mathbb{R}_{>1}$. Further, the run-time can be improved when the positive degree of the graph is bounded, as shown in the following corollary.

Corollary 1. Let Δ denote the maximum positive degree in an instance $G = (V, E)$ of unweighted, complete correlation clustering on $|V| = n$ nodes. Suppose G is given as an adjacency list representation of its positive edges. There exists a combinatorial algorithm returning a single clustering that is simultaneously an $O(1)$-approximation for all ℓ_p-norm objectives, for all $p \in \mathbb{R}_{\geqslant 1} \cup \{\infty\}$, and its run-time is $O(n\Delta^2 \log n)$.

[4]In some of the literature, for instance that of Golovin et al. [23], it is called the all-ℓ_p-norms objective.

[5]Note this is independent of p.

The run-time of the algorithm matches the fastest known algorithm for the ℓ_∞-norm objective [19], in both the general case and when the maximum positive degree is bounded. The best-known algorithm before our work relied on solving a convex relaxation on $|V|^2$ variables and $|V|^3$ constraints. We improve the run-time by avoiding this bottleneck.

In the setting when the positive edges form a regular graph, the interested reader may also find a rather beautiful proof (which is much simpler than that of Theorem 1) in Section 3 showing there is a solution that is simultaneously $O(1)$-approximate for the ℓ_1-norm and ℓ_∞-norm objectives.

1.2 Related work

Correlation clustering was introduced by Bansal, Blum, and Chawla [2004]. The version they introduced also studies the problem on unweighted, complete graphs, but is concerned with minimizing the ℓ_1-norm of the disagreement vector. For this problem, Ailon, Charikar, and Newman [2008] designed the Pivot algorithm, which is a randomized algorithm that in expectation obtains a 3-approximation. While we know algorithms with better approximations for ℓ_1 correlation clustering than Pivot [16, 18], the algorithm remains a baseline in correlation clustering due to its simplicity. (However, Pivot can perform arbitrarily badly—i.e., give $\Omega(n)$ approximation ratios—for other ℓ_p-norms; see again the example in Appendix A of [34].) It is an active area of research to develop algorithms for the ℓ_1-norm that focus on practical scalability [11, 17, 32, 35, 14]. Correlation clustering has also been studied on non-complete, weighted graphs [15, 28], with conditions on the cluster sizes [33], and with asymmetric errors [25]. In fact, in recent work Veldt [36] highlighted the need for deterministic techniques in correlation clustering that do not use linear programming. Much interest in correlation clustering stems from its connections to applications, including community detection, natural language processing, location area planning, and gene expression [37, 35, 38, 31, 9, 20].

Puleo and Milenkovic [2016] introduced correlation clustering with the goal of minimizing the ℓ_p-norm of the disagreement vector. They show that even for minimizing the ℓ_∞-norm on complete, unweighted graphs, the problem is NP-hard (Appendix C in [34]). Several groups found $O(1)$-approximation algorithms for minimizing the ℓ_p-norm on complete, unweighted graphs [34, 15, 28], the best of which is currently the 5-approximation of Kalhan, Makarychev, and Zhou [2019]. Many other interesting objectives for correlation clustering focus on finding solutions that are (in some sense) fair or locally desirable [3, 8, 2, 21, 26, 1]. All of these previous works that study general ℓ_p-norms or other notions of fairness or locality rely on solving a convex relaxation. This has two downsides: (1) the run-time of the algorithms are bottle-necked by the time it takes to solve the relaxation with at least $\Omega(n^2)$ many variables and $\Omega(n^3)$ constraints; in fact, it is time-consuming to even enumerate the $\Omega(n^2)$ variables and $\Omega(n^3)$ constraints; and (2) the solution is only guaranteed to be good for one particular value of p.

Several problems have been studied with the goal of finding a solution that is a good approximation for several objectives simultaneously. The all-norms objective was introduced by Azar et al. [6], where the goal is to design a ρ-approximation algorithm for all ℓ_p-norm objectives of a problem. They originally introduced the objective for the restricted assignment load balancing problem and showed an all-norms 2-approximation. Further follow-up on the all-norms objective has been done for load balancing [29, 10, 30], and for set cover [23]. The term "universal" algorithm has also been used for Steiner tree [12, 13], TSP [27], and clustering [22], though in these settings the goal is different, namely, to find a solution that is good for any potential input; e.g., in Universal Steiner Tree, the goal is to find a spanning tree where for any set of terminals, the sub-tree connecting the root to the terminals is a good approximation of the optimal.

2 Preliminaries

We will introduce notation, and then we will discuss two relevant works—the papers by Kalhan, Makarychev, and Zhou [2019] and Davies, Moseley, and Newman [2023].

2.1 Notation

Recall our input to the correlation clustering problem is $G = (V, E)$, an unweighted, complete graph on n vertices, and every edge is assigned a label of either positive ($+$) or negative ($-$). Let the set of positive edges be denoted E^+ and the set of negative edges E^-. Then, we can define the positive neighborhood and negative neighborhood of a vertex u as $N_u^+ = \{v \in V \mid (u, v) \in E^+\}$ and $N_u^- = \{v \in V \mid (u, v) \in E^-\}$, respectively. We further assume without loss of generality that every vertex has a positive self-loop to itself.

A clustering $\mathcal{C}$ is a partition of V into clusters $C_1, \ldots, C_k$ (but recall that k is not pre-specified). Let $C(u)$ denote the cluster that vertex u is in, i.e., if $\mathcal{C}$ has k clusters, there exists exactly one $i \in [k]$ such that $C(u) = C_i$. It is also helpful to consider the vertices in a different cluster than u, and so we let $\overline{C(u)} = V \backslash C(u)$ denote this. We say that a positive edge $e = (u, v) \in E^+$ is a disagreement with respect to $\mathcal{C}$ if $v \in \overline{C(u)}$. On the other hand, we say that a negative edge $e = (u, v) \in E^-$ is a disagreement with respect to $\mathcal{C}$ if $v \in C(u)$. For a fixed clustering $\mathcal{C}$, we denote the disagreement vector of $\mathcal{C}$ as $y_{\mathcal{C}} \in \mathbb{Z}^n_{\geqslant 0}$, where for $u \in V$, $y_{\mathcal{C}}(u)$ is the number of edges incident to u that are disagreements with respect to $\mathcal{C}$. We omit the subscript throughout the proofs when a clustering is clear.

Throughout, we let OPT be the optimal objective value, and the ℓ_p-norm to which it corresponds will be clear from context. The next fact follows from the definitions seen so far (recalling also the positive self-loops).

Fact 1. For any $u, v \in V$, possibly with $u = v$, $n = |N_u^+ \cap N_v^+| + |N_u^- \cap N_v^-| + |N_u^+ \cap N_v^-| + |N_u^- \cap N_v^+|$.

2.2 Summary of work by Kalhan, Makarychev, and Zhou

The standard linear program relaxation for correlation clustering is stated in LP 2.2.[6] In the integer LP, the variable x_{uv} indicates whether vertices u and v will be in the same cluster (0 for yes, 1 for no), and the disagreement vector is y; the optimal solution to the integer LP has value OPT, while the optimal solution to the relaxation gives a lower bound on OPT. Note the triangle inequality is enforced on all triples of vertices, inducing a semi-metric space on V. Throughout this paper, as in [19], we refer to the algorithm by Kalhan, Makarychev, and Zhou as the KMZ algorithm. The KMZ algorithm has two phases: it solves LP 2.2, and then uses the KMZ rounding algorithm (Algorithm 1) to obtain an integral assignment of vertices to clusters. See Appendix D for its formal statement. At a high-level, the KMZ rounding algorithm is an iterative, ball-growing algorithm that uses the semi-metric to guide its choices on forming clusters. Their algorithm is a 5-approximation, and produces different clusterings for different p, since the optimal solution x^* to LP 2.2 depends on p.

LP 2.2

$$\begin{aligned} & \min \|y\|_p \\ \text{s.t. } & y_u = \sum_{v \in N_u^+} x_{uv} + \sum_{v \in N_u^-} (1 - x_{uv}) && \forall u \in V \end{aligned}$$

[6] Technically this is a convex program as the objective is convex. For simplicity we will refer to it as an LP as the constraints are linear.

$$x_{uv} \leqslant x_{uw} + x_{vw} \qquad \forall u, v, w \in V$$
$$0 \leqslant x_{uv} \leqslant 1 \qquad \forall u, v \in V.$$

Definition 1. Let f be a semi-metric on V, i.e., taking $x = f$ gives a feasible solution to LP 2.2. The fractional cost of f in the ℓ_p-norm objective is the value of LP 2.2 that results from setting $x = f$. When p is clear from context, we will simply call this the fractional cost of f.

2.3 Summary of work by Davies, Moseley, and Newman

The main take-away from the work of Kalhan, Makarychev, and Zhou [2019] is that one only requires a semi-metric on the set of vertices, whose cost is comparable to the cost of an optimal solution, as input to the KMZ rounding algorithm (Algorithm 1). Thus, the insight of Davies, Moseley, and Newman [2023] for the ℓ_∞-norm objective is that one can combinatorially construct such a semi-metric without solving an LP, and at small loss in the quality of the fractional solution. They do this by introducing the correlation metric.

Definition 2 ([19]). For all $u, v \in V$, the correlation metric defines the distance between u and v as

$$\begin{aligned} d_{uv} &= 1 - \frac{|N_u^+ \cap N_v^+|}{|N_u^+ \cup N_v^+|} \\ &= \frac{|N_u^+ \cap N_v^-| + |N_u^- \cap N_v^+|}{|N_u^+ \cap N_v^+| + |N_u^+ \cap N_v^-| + |N_u^- \cap N_v^+|}. \end{aligned} \tag{1}$$

Note that the rewrite in Line (1) is apparent from Fact 1.

The correlation metric captures useful information succinctly. Intuitively, if u and v have relatively large positive intersection, i.e., $N_u^+ \cap N_v^+$ is large compared to their other relevant joint neighborhoods $(N_u^+ \cap N_v^-) \cup (N_u^- \cap N_v^+)$, then from the perspective of u and v, fewer disagreements are incurred by putting u and v in the same cluster than by putting them in different clusters. This is because if u and v are in the same cluster, then they have disagreements on edges (u, w) and (v, w) for $w \in (N_u^+ \cap N_v^-) \cup (N_u^- \cap N_v^+)$, but if they are in different clusters, then u and v have disagreements on edges (u, w) and (v, w) for $w \in N_u^+ \cap N_v^+$. Note that the metric is not normalized by n (the size of the joint neighborhood), but instead by $n - |N_u^- \cap N_v^-| = |N_u^+ \cap N_v^+| + |N_u^+ \cap N_v^-| + |N_u^- \cap N_v^+|$. In addition to the above intuition on the normalization factor, we also observe that $w \in N_u^- \cap N_v^-$ do not necessarily force disagreements on (u, w) and (v, w), since w can go in a different cluster than both u and v without penalty. For more on intuition behind the correlation metric, see Section 2 in [19].

Davies, Moseley, and Newman [2023] prove that the correlation metric d can be used as input to the KMZ rounding algorithm by showing that (1) d satisfies the triangle inequality and (2) the fractional cost of d in the ℓ_∞-norm (recall Definition 1) is no more than 8 times the value of the optimal integral solution (OPT). Since the KMZ rounding algorithm (Algorithm 1) loses a factor of at most 5, inputting d to that algorithm returns a 40-approximation algorithm. A benefit of the correlation metric is that it can be computed in time $O(n^\omega)$, and even faster when the subgraph on positive edges is sparse.

2.4 Technical overview

It is not hard to see that the correlation metric cannot be used as input to the KMZ algorithm for ℓ_p-norms other than $p = \infty$, as one cannot bound the fractional cost of the correlation metric against the optimal with only an $O(1)$-factor loss. To see why, consider the star again, as in Figure

1. Here, for all $u, v \in \{v_1, \ldots, v_{n-1}\}$, $d_{uv} = 1 - 1/(n-(n-3)) = 2/3$, but for the ℓ_1-norm, we need the semi-metric to have the value $1 - d_{uv}$ be close to 0, i.e. $O(1/n)$, for such u, v, in order for the fractional cost to be comparable to the value of OPT for $p = 1$.

There are several possible fixes one could try to make to the correlation metric. One idea is that since one can interpret the correlation metric as a coarse approximation of the probability the Pivot algorithm separates u and v[7], one could try to adapt the correlation metric to more accurately approximate this probability.[8] Another idea, inspired by an observation below, is that one could define a semi-metric for edges in E^+ and another semi-metric for edges in E^-, but then there is the difficulty of showing the triangle inequality holds when positive and negative edges are mixed together. Both of these ideas were, for us, unsuccessful.

Instead, the following two observations of how the correlation metric works with respect to the ℓ_1-norm led us to an effective adaptation:

1. One can bound the fractional cost, restricted to positive edges, of the correlation metric in the ℓ_1-norm by an $O(1)$-factor times the optimal solution's cost (see Claim 1 in Appendix C). Negative edges still pose a challenge.

2. If the subgraph of positive edges is regular, then we show one can bound the fractional cost of the correlation metric in the ℓ_1-norm on negative edges as well as positive.[9] See Section 3.

These observations led us to ask whether some adjustments to the correlation metric might yield a semi-metric with bounded fractional cost in the ℓ_1-norm or even the ℓ_p-norm more generally (while still remaining bounded in the ℓ_∞-norm). Moreover, since the KMZ rounding algorithm does not depend on p (whereas in the KMZ algorithm, the solution to the LP does depend on p), inputting the same semi-metric to the rounding algorithm produces the same clustering for all ℓ_p-norms!

We are ready to define the adjusted correlation metric. Let Δ_u denote the positive degree of u (the degree of u in the subgraph of positive edges).

Definition 3. Define the adjusted correlation metric $f : E \to [0, 1]$ as follows:

1. For d the correlation metric, i.e., $d_{uv} = 1 - \frac{|N_u^+ \cap N_v^+|}{|N_u^+ \cup N_v^+|}$, initially set $f = d$.

2. If $e \in E^-$ and $d_e > 0.7$, set $f_e = 1$ (round up).

3. For $u \in V$ such that $|N_u^- \cap \{v : d_{uv} \leqslant 0.7\}| \geqslant \frac{10}{3}\Delta_u$, set $f_{uv} = 1$ for all $v \in V \backslash \{u\}$.

The idea in Step 3 is that if the fractional cost of negative edges incident to u is sufficiently large, we instead trade this for the cost of positive disagreements, as the rounding algorithm will now put u in its own cluster. For the ℓ_∞-norm, this trade-off is innocuous. For ℓ_p-norms in general, a refined charging argument is needed to show that post-processing d in this way sufficiently curbs the (too large) fractional cost of d.

In Section 3, we start with a warm-up exercise and show that if the graph on positive edges is regular, then the (original) correlation metric d has $O(1)$-approximate fractional cost. Note this section is not necessary to understanding the rest of the paper, but we include it in the main body

[7]Pivot operates as follows: Choose a random unclustered $u \in V$. Take u and all its unclustered positive neighbors and let this be the newest cluster. Continue until all vertices in V are clustered.

[8]If in LP 2.2, x_{uv} is set exactly to the probability that u and v are separated by Pivot, then x will be a feasible solution with cost at most $3\mathsf{OPT}$. However, this probability seems difficult to express in closed form, even approximately.

[9]In contrast, one cannot bound the fractional cost of the correlation metric on the (irregular) star (Figure 1).

because we find the proof here very clever! The main technical result of the paper is Section 4, where we prove Theorem 1 by showing that the adjusted correlation metric can be input to the KMZ rounding algorithm. Namely, we will first show (quite easily) that the adjusted correlation metric satisfies an approximate triangle inequality. Then, it remains to upper bound the fractional cost of the adjusted correlation metric against OPT. We tackle this with a combinatorial charging argument. This argument leverages a somewhat different approach from that used in [19] and is simpler than their proof for only the ℓ_∞-norm. The constant approximation factor obtained from inputting the adjusted correlation metric to Algorithm 1 is bounded above (and below) by universal constants for all p (this is the worst case and one can get better constants for each p). While we keep the argument for general ℓ_p-norms in the main body of the paper, the interested reader may find a simplified proof for the ℓ_1-norm objective in Appendix C.

3 A Special Case: Regular Graphs

In general, the original correlation metric d (Definition 2) does not necessarily have bounded fractional cost for the ℓ_1-norm objective (or more generally for ℓ_p-norm objectives). So, we use the adjusted correlation metric f (Definition 3) as input to the KMZ rounding algorithm (Appendix D). In this section, we show that if the subgraph of positive edges is regular, then the correlation metric d can be used as is (i.e., without the adjustments in Steps 2 and 3 of Definition 3) to yield a clustering that is constant approximate for the ℓ_1-norm and ℓ_∞-norm simultaneously:

Theorem 2. Let $G = (V, E)$ be an instance of unweighted, complete correlation clustering, and let E^+ denote the set of positive edges. Suppose that the subgraph induced by E^+ is regular. Then the fractional cost of d in the ℓ_1-norm objective is within a constant factor of OPT:

$$\sum_{u \in V} \sum_{v \in N_u^+} d_{uv} + \sum_{u \in V} \sum_{v \in N_u^-} (1 - d_{uv}) = O(\mathit{OPT}).$$

Therefore, the clustering produced by inputting d to the KMZ rounding algorithm is a constant-factor approximation simultaneously for the ℓ_1-norm and ℓ_∞-norm objectives.

While this proof is quite different than the one for arbitrary-degree graphs, we include it because we find it beautiful, and it gives useful intuition for where the correlation metric breaks down for irregular graphs.

Proof. Let Δ be the (common) degree of the positive subgraph. To show that the fractional cost of d in the ℓ_1-norm objective is $O(\mathsf{OPT})$ for regular graphs, we will use a dual fitting argument. The LP relaxation we consider is from [4], which uses a dual fitting argument to show constant approximation guarantees for Pivot (although the proof here does not otherwise resemble the proof for Pivot). The primal is given by

$$\min \Big\{ \sum_{e \in E} x_e \mid x_{ij} + x_{jk} + x_{ki} \geqslant 1, \forall ijk \in \mathcal{T}, x \geqslant 0 \Big\} \tag{P}$$

where $\mathcal{T}$ is the set of bad triangles (i.e. triangles with exactly two positive edges and one negative edge). For $x \in \{0,1\}^{|E|}$, x corresponds to disagreements in a clustering: we set $x_e = 1$ if e is a disagreement and $x_e = 0$ otherwise. The constraints state that every clustering must make a disagreement on every bad triangle. Thus, (P) is a relaxation for the ℓ_1-norm objective. In fact, we will prove the stronger statement that the fractional cost is $O(\mathsf{OPT}_P)$, where OPT_P is the optimal objective value of (P).

The dual is given by

$$\max\Big\{ \sum_{T\in\mathcal{T}} y_T \mid \sum_{T\in\mathcal{T}:T\ni e} y_T \leqslant 1, \forall e\in E, y\geqslant 0\Big\}. \tag{D}$$

We show that by setting $y_T = \frac{1}{2\Delta}$ for all $T\in\mathcal{T}$, y satisfies the following properties:

1. y is feasible in (D).
2. The fractional cost of d is at most $6\cdot\sum_{T\in\mathcal{T}} y_T$.

Letting OPT_D be the optimal objective value of (D), we have $6\cdot\sum_{T\in\mathcal{T}} y_T \leqslant 6\cdot\mathsf{OPT}_D = 6\cdot\mathsf{OPT}_P \leqslant 6\cdot\mathsf{OPT}$, which will conclude the proof.

To prove feasibility, we case on whether e is positive or negative.

If $e\in E^-$, then $|\{T\in\mathcal{T} : T\ni e\}| = |N_u^+\cap N_v^+| \leqslant \Delta$, where equality is by the definition of a bad triangle. So

$$\sum_{T\in\mathcal{T}:T\ni e} y_T \leqslant \frac{\Delta}{2\Delta} \leqslant 1.$$

If $e\in E^+$, then $|\{T : T\ni e\}| = |(N_u^+\cap N_v^-)\cup(N_u^-\cap N_v^+)| \leqslant 2\Delta$. We conclude y is feasible, since

$$\sum_{T\in\mathcal{T}:T\ni e} y_T \leqslant \frac{2\Delta}{2\Delta} = 1.$$

Now we need to show that the fractional cost of d is bounded in terms of the objective value of (D). First we bound the fractional cost of the negative edges:

$$\sum_{(u,v)\in E^-}(1-d_{uv}) \leqslant \sum_{(u,v)\in E^-}|N_u^+\cap N_v^+|/\Delta = \sum_{e\in E^-}\sum_{T\in\mathcal{T}:T\ni e} 1/\Delta = \sum_{e\in E^-}\sum_{T\in\mathcal{T}:T\ni e} 2y_T,$$

where in the first inequality we have used that $|N_u^+\cup N_v^+|\geqslant\Delta$. Next we bound the fractional cost of the positive edges:

$$\sum_{(u,v)\in E^+} d_{uv} \leqslant \sum_{(u,v)\in E^+}(|N_u^+\cap N_v^-| + |N_u^-\cap N_v^+|)/\Delta = \sum_{e\in E^+}\sum_{T\in\mathcal{T}:T\ni e} 1/\Delta = \sum_{e\in E^+}\sum_{T\in\mathcal{T}:T\ni e} 2y_T.$$

So the total fractional cost is bounded by $\sum_{e\in E}\sum_{T\in\mathcal{T}:T\ni e} 2y_T = 6\cdot\sum_{T\in\mathcal{T}} y_T$, since each triangle contains three edges. This is what we sought to show. Since the fractional cost of d is bounded for the ℓ_1-norm objective (and the ℓ_∞-norm objective by [19]), using d as input to KMZ rounding algorithm produces a clustering that is simultaneously $O(1)$-approximate for the ℓ_1- and ℓ_∞-norm objectives. □

4 Proof of Theorem 1

The goal of this section is to prove Theorem 1 and the subsequent Corollary 1. We begin by outlining that the adjusted correlation metric satisfies an approximate triangle inequality in Subsection 4.1. Then in Subsection 4.2, we prove the fractional cost of the adjusted correlation metric in any ℓ_p-norm objective is an $O(1)$ factor away from the optimal solution's value. We tie it all together to prove Theorem 1 and Corollary 1 in Subsection 4.3.

Before all that, we prove a proposition that will be key in several settings. Loosely, it states that if two vertices are close to each other according to d, then they have a large shared positive neighborhood.

Proposition 1. Fix vertices $u, v \in V$ and a clustering $\mathcal{C}$ on V such that $d_{uv} \leqslant 0.7$ and $|N_u^+ \cap C(u)| \,/\, |N_u^+| \geqslant 0.85$. Then $|N_u^+ \cap N_v^+ \cap C(u)| \geqslant 0.15 \cdot |N_u^+|$.

Proof. Since $d_{uv} \leqslant 0.7$, $\frac{|N_u^+ \cap N_v^+|}{|N_u^+|} \geqslant \frac{|N_u^+ \cap N_v^+|}{|N_u^+ \cup N_v^+|} \geqslant 0.3$. Using the assumption on u together with the above inequality implies that $|N_v^+ \cap N_u^+ \cap C(u)| \,/\, |N_u^+| \geqslant 0.85 + 0.3 - 1 = 0.15$. □

4.1 Triangle inequality

Recall that the correlation metric d satisfies the triangle inequality (see Section 4.2 in [19]). We will show that the adjusted correlation metric f satisfies an approximate triangle inequality, which is sufficient for the KMZ rounding algorithm. Formally, we say that a function g is a δ-semi-metric on some set S if it is a semi-metric on S, except instead of satisfying the triangle inequality, g satisfies $g(u, v) \leqslant \delta \cdot (g(u, w) + g(v, w))$ for all $u, v, w \in S$.

Lemma 1 (Triangle Inequality). The adjusted correlation metric f is a $\frac{10}{7}$-semi-metric.

The proof of Lemma 1 is straightforward given that d satisfies the triangle inequality; see Appendix A.

Lemma 3 in [19] proves that one can input a semi-metric that satisfies an approximate triangle inequality (instead of the exact triangle inequality) to the KMZ rounding algorithm (with some loss in the approximation factor). We summarize the main take-away below.

Lemma 2 ([19]). If g is a δ-semi-metric on the set V, instead of a true semi-metric (i.e., 1-semi-metric), then the KMZ algorithm loses a factor of $1 + \delta + \delta^2 + \delta^3 + \delta^4$.[10]

Since we show in Lemma 1 that f is a $\frac{10}{7}$-semi-metric, we lose a factor of 12 in inputting f to the KMZ algorithm (along with the factor loss from the fractional cost).

4.2 Bounding the fractional cost of ℓ_p-norms

This section bounds the fractional cost of the adjusted correlation metric for the ℓ_p-norms. The following lemma considers the case where $p = \infty$. The general case is handled in the subsequent lemma.

Lemma 3. The fractional cost of the adjusted correlation metric f in the ℓ_∞-norm objective is at most $56 \cdot \mathit{OPT}$, where OPT is the cost of the optimal integral solution in the ℓ_∞-norm.

The lemma follows from the fact that the fractional cost of the correlation metric d in the ℓ_∞-norm is known to be bounded by [19], and that it only decreases when d is replaced by f due to Definition 3. For completeness, we include a proof in Appendix B.

For finite $p \in \mathbb{R}_{\geqslant 1}$, our techniques hold for any ℓ_p-norm, as we only require the convexity of x^p. As far as we know, there has been no study initiated on the problem for $0 \leqslant p < 1$. We use two primary lemmas—one for the positive edge fractional cost and one for the negative edge fractional cost—to show that the adjusted correlation metric well approximates the optimal for general ℓ_p-norms.

Lemma 4. The fractional cost of the adjusted correlation metric f in the ℓ_p-norm objective is a constant factor (independent of p) away from the cost of the optimal integral solution in the ℓ_p-norm.

[10] When $\delta = 1$, this factor equals 5, which is the loss in the KMZ algorithm.

Proof. Let y be the disagreement vector for an optimal clustering $\mathcal{C}$ in the ℓ_p-norm, for any $p \in \mathbb{R}_{\geqslant 1} \cup \{\infty\}$. When $p = \infty$, see Lemma 3. For $p \in \mathbb{R}_{\geqslant 1}$, by definition $\mathsf{OPT}^p = \sum_{w \in V} (y(w))^p$, and the pth power of the fractional cost of f is given by

$$cost(f)^p = \sum_{u \in V} \Big[\sum_{v \in N_u^+} f_{uv} + \sum_{v \in N_u^-} (1 - f_{uv}) \Big]^p.$$

$$\text{Observe that} \quad cost(f)^p \leqslant 2^p \underbrace{\sum_{u \in V} \Big(\sum_{v \in N_u^+} f_{uv} \Big)^p}_{(S^+)^p} + 2^p \underbrace{\sum_{u \in V} \Big(\sum_{v \in N_u^-} (1 - f_{uv}) \Big)^p}_{(S^-)^p}.$$

We refer to bounding $(S^+)^p$ as bounding the fractional cost of the positive edges, and likewise $(S^-)^p$ for the negative edges. The first sum, $(S^+)^p$, is bounded in Lemma 5 and the second sum, $(S^-)^p$, is bounded in Lemma 6. Using those two bounds, together we have $cost(f) \leqslant [2^p((S^+)^p + (S^-)^p)]^{1/p} \leqslant 529$, for $p \in [1, \infty)$. Specifically, the middle term is maximized at $p = 1$, giving the bound of 529, and tends to below 214 as $p \to \infty$. (A more tailored analysis in Appendix C gives a constant of 74 for $p = 1$.) □

We note that, as our main interest is determining whether a simultaneous constant approximation is even possible (and a combinatorial one, at that), we did not pay particular attention to optimizing constants, but suspect these could be greatly reduced.

4.2.1 Fractional cost of positive edges in ℓ_p-norms

Bounding the fractional cost of the positive edges in the ℓ_p-norm will be similar to in the simpler ℓ_1-norm (see Appendix C), only there will be an extra step in which we apply Jensen's inequality.

Lemma 5. For $p \in \mathbb{R}_{\geqslant 1}$, the fractional cost of the adjusted correlation metric f in the ℓ_p-norm objective for the set of positive edges is a constant factor approximation to the optimal, i.e.,

$$(S^+)^p = \sum_{u \in V} \Big(\sum_{v \in N_u^+} f_{uv} \Big)^p \leqslant 2^p \cdot [(8^p/2 + 1)((20/3)^p + 2 + 2 \cdot 4^p) + 8^p + 1] \cdot \mathsf{OPT}^p.$$

One of the challenges in bounding the cost of f is that disagreements in the ℓ_p-norm objective for $p \neq 1$ are asymmetric, in that a disagreeing edge charges $y(u)$ and $y(v)$ (whereas for $p = 1$ we can just sum the number of disagreeing edges). Step 3 rounds up the edges incident to u when the tradeoff is good from u's perspective. However, an edge (u, v) may be rounded up to 1 when this tradeoff is good from v's perspective, but not from u's perspective. The high-level idea for why this is fine is that if u and v are close under d, their positive neighborhoods overlap significantly and, in some average sense, u can charge to v. Proving this requires a double counting argument using a bipartite auxiliary graph. If u and v are far under d, on the other hand, we can charge to the cost of the correlation metric, which will be bounded on an appropriate subgraph.

The second challenge is showing that the ℓ_p-norm of the disagreement vector, restricted to vertices u that are made singletons in Step 3, is bounded. This again requires a double counting argument.

Proof. Fix an optimal clustering $\mathcal{C}$. We partition vertices based on membership in $C(u)$ or $\overline{C(u)}$ (as defined in Subsection 2.1). Let y denote the disagreement vector of $\mathcal{C}$. We have

$$(S^+)^p = \sum_{u \in V} \Big(\sum_{v \in N_u^+} f_{uv} \Big)^p \leqslant 2^p \underbrace{\sum_{u \in V} \Big(\sum_{v \in N_u^+ \cap C(u)} f_{uv} \Big)^p}_{S_1^+} + 2^p \underbrace{\sum_{u \in V} \Big(\sum_{v \in N_u^+ \cap \overline{C(u)}} f_{uv} \Big)^p}_{S_2^+}.$$

It is easy to bound S_2^+ by using the trivial upper bound $f_{uv} \leqslant 1$:

$$\boxed{S_2^+ = \sum_{u \in V} \Big(\sum_{v \in N_u^+ \cap \overline{C(u)}} f_{uv} \Big)^p \leqslant \sum_{u \in V} \Big(\sum_{v \in N_u^+ \cap \overline{C(u)}} 1 \Big)^p \leqslant \sum_{u \in V} (y(u))^p = \mathsf{OPT}^p,}$$

where we have used that every edge $(u, v) \in E^+$ with $v \notin C(u)$ is a disagreement incident to u.

Next, we bound S_1^+. Let R_1 be the set of u for which Step 3 of Definition 3 applies. For these u, we have $f_{uv} = 1$ for all $v \in V \backslash \{u\}$. Let $R_2 = V \backslash R_1$. For $u \in R_2$ and $v \in N_u^+$, we have that either $v \in R_2$, in which case $f_{uv} = d_{uv}$; or $v \in R_1$, in which case $f_{uv} = 1$. (Note that V is the disjoint union of R_1 and R_2.) So

$$S_1^+ = \underbrace{\sum_{u \in R_1} \Big(\sum_{v \in N_u^+ \cap C(u), v \neq u} 1 \Big)^p}_{S_{11}^+} + \underbrace{\sum_{u \in R_2} \Big(\sum_{v \in N_u^+ \cap C(u)} f_{uv} \Big)^p}_{S_{12}^+},$$

and in particular

$$\begin{aligned}
S_{12}^+ &= \sum_{u \in R_2} \Big(\sum_{v \in N_u^+ \cap C(u) \cap R_1} 1 + \sum_{v \in N_u^+ \cap C(u) \cap R_2} d_{uv} \Big)^p \\
&= \sum_{u \in R_2} \Big(\sum_{\substack{v \in N_u^+ \cap C(u) \cap R_1 \\ d_{uv} \leqslant 1/4}} 1 + \sum_{\substack{v \in N_u^+ \cap C(u) \cap R_1 \\ d_{uv} \geqslant 1/4}} 1 + \sum_{v \in N_u^+ \cap C(u) \cap R_2} d_{uv} \Big)^p \\
&\leqslant \sum_{u \in R_2} \Big(\sum_{\substack{v \in N_u^+ \cap C(u) \cap R_1 \\ d_{uv} \leqslant 1/4}} 1 + \sum_{\substack{v \in N_u^+ \cap C(u) \cap R_1 \\ d_{uv} \geqslant 1/4}} 4 \cdot d_{uv} + \sum_{v \in N_u^+ \cap C(u) \cap R_2} d_{uv} \Big)^p \\
&\leqslant \sum_{u \in R_2} \Big(\sum_{\substack{v \in N_u^+ \cap C(u) \cap R_1 \\ d_{uv} \leqslant 1/4}} 1 + \sum_{v \in N_u^+ \cap C(u)} 4 \cdot d_{uv} \Big)^p \leqslant 2^p \underbrace{\sum_{u \in R_2} \Big(\sum_{\substack{v \in N_u^+ \cap R_1 \\ d_{uv} \leqslant 1/4}} 1 \Big)^p}_{S_{13}^+} + 8^p \cdot \underbrace{\sum_{u \in R_2} \Big(\sum_{v \in N_u^+ \cap C(u)} d_{uv} \Big)^p}_{S_{14}^+}.
\end{aligned}$$

First we will bound S_{13}^+. To do so, we will strongly use that $d_{uv} \leqslant 1/4$ in the inner sum. In particular, we will make use of the following easy proposition.

Proposition 2. Let d be the correlation metric, and suppose $d_{uv} \leqslant 1/4$. Then $|N_u^+| \leqslant \frac{7}{3} \cdot |N_v^+|$.

Proof of Proposition 2. Since $d_{uv} \leqslant 1/4$ implies $1 - d_{uv} \geqslant 3/4$, we see that

$$\frac{|N_u^+ \cap N_v^+|}{|N_u^+| + |N_v^+| - |N_u^+ \cap N_v^+|} \geqslant 3/4,$$

and we complete the proof since $7 \cdot |N_v^+| \geqslant 7 \cdot |N_u^+ \cap N_v^+| \geqslant 3 \cdot |N_u^+| + 3 \cdot |N_v^+| \geqslant 3 \cdot |N_u^+|$. □

Next, we will need to create a bipartite auxiliary graph $H = (R_2, R_1, F)$ with R_2 and R_1 being the sides of the partition, and F being the edge set. We will then use a double counting argument. Place an edge between $u \in R_2$ and $v \in R_1$ if $uv \in E^+$ and $d_{uv} \leqslant 1/4$. Then we have precisely that

$$S_{13}^+ = \sum_{u \in R_2} \deg_H(u)^p.$$

We will show that

$$\boxed{S_{13}^+ = \sum_{u \in R_2} \deg_H(u)^p \leqslant 4^{p-1} \cdot \sum_{v \in R_1} |N_v^+|^p \leqslant 4^{p-1} \cdot ((20/3)^p + 2 + 2 \cdot 4^p) \cdot \mathsf{OPT}^p} \tag{2}$$

where the last bound follows from Proposition 3, which we establish separately below. We will bound via double counting the quantity L, defined below. Let $N_H(\cdot)$ denote the neighborhoods in H of the vertices.

$$\begin{aligned} L &:= \sum_{f=uv \in F} (\deg_H(u) + \deg_H(v))^{p-1} \leqslant \sum_{v \in R_1} \sum_{u \in N_H(v)} (\deg_H(v) + \deg_H(u))^{p-1} \\ &\leqslant \sum_{v \in R_1} \sum_{u \in N_H(v)} \left(|N_v^+| + |N_u^+|\right)^{p-1} \leqslant \sum_{v \in R_1} \sum_{u \in N_H(v)} 4^{p-1} \cdot |N_v^+|^{p-1} \qquad (3) \\ &\leqslant 4^{p-1} \cdot \sum_{v \in R_1} |N_v^+| \cdot |N_v^+|^{p-1} = 4^{p-1} \cdot \sum_{v \in R_1} |N_v^+|^p \end{aligned}$$

where in (3) we've used Proposition 2. Note that L is upper bounded by the right-hand side in (2). Now it just remains to show that L is lower bounded by the left-hand side in (2).

$$\begin{aligned} L &= \sum_{f=uv \in F} (\deg_H(u) + \deg_H(v))^{p-1} = \sum_{u \in R_2} \sum_{v \in N_H(u)} (\deg_H(u) + \deg_H(v))^{p-1} \\ &\geqslant \sum_{u \in R_2} \sum_{v \in N_H(u)} \deg_H(u)^{p-1} = \sum_{u \in R_2} \deg_H(u) \cdot \deg_H(u)^{p-1} = \sum_{u \in R_2} \deg_H(u)^p, \end{aligned}$$

which is what we sought to show. Now we bound S_{14}^+.

$$\begin{aligned} S_{14}^+ &\leqslant \sum_{u \in V} \left(\sum_{v \in N_u^+ \cap C(u)} \frac{|N_u^+ \cap N_v^-| + |N_u^- \cap N_v^+|}{|N_u^+ \cup N_v^+|} \right)^p \\ &\leqslant \sum_{u \in V} \left(\sum_{v \in N_u^+} \frac{y(u) + y(v)}{|N_u^+ \cup N_v^+|} \right)^p \leqslant \sum_{u \in V} |N_u^+|^{p-1} \sum_{v \in N_u^+} \frac{(y(u) + y(v))^p}{|N_u^+ \cup N_v^+|^p} \\ &\leqslant 2^p \sum_{u \in V} \sum_{v \in N_u^+} |N_u^+|^{p-1} \cdot \frac{y(u)^p}{|N_u^+ \cup N_v^+|^p} + 2^p \sum_{u \in V} \sum_{v \in N_u^+} |N_u^+|^{p-1} \cdot \frac{y(v)^p}{|N_u^+ \cup N_v^+|^p}. \end{aligned}$$

In the second line, the first inequality uses the fact that for $w \in (N_u^+ \cap N_v^-) \cup (N_u^- \cap N_v^+)$, then at least one of $(u, w), (v, w)$ is a disagreement, since $v \in C(u)$ in the inner summation of the first line. The second inequality in the second line uses Jensen's inequality.

To bound the first double sum above, we use an averaging argument:

$$\sum_{u \in V} \sum_{v \in N_u^+} |N_u^+|^{p-1} \cdot \frac{y(u)^p}{|N_u^+ \cup N_v^+|^p} \leqslant \sum_{u \in V} \sum_{v \in N_u^+} \frac{y(u)^p}{|N_u^+|} = \sum_{u \in V} y(u)^p = \mathsf{OPT}^p.$$

To bound the second double sum, we first have to flip it:

$$\begin{aligned} \sum_{u \in V} \sum_{v \in N_u^+} |N_u^+|^{p-1} \cdot \frac{y(v)^p}{|N_u^+ \cup N_v^+|^p} &= \sum_{v \in V} \sum_{u \in N_v^+} |N_u^+|^{p-1} \cdot \frac{y(v)^p}{|N_u^+ \cup N_v^+|^p} \\ &\leqslant \sum_{v \in V} \sum_{u \in N_v^+} |N_u^+|^{p-1} \frac{y(v)^p}{|N_u^+|^{p-1} \cdot |N_v^+|} = \sum_{v \in V} y(v)^p = \mathsf{OPT}^p. \end{aligned}$$

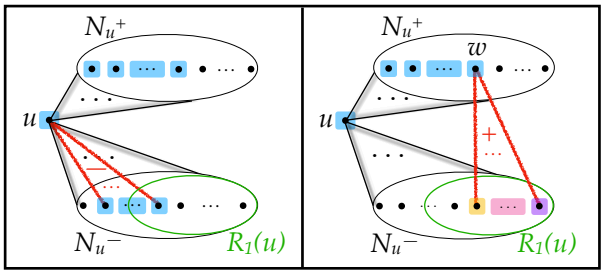


Figure 2: Left: Case 2a(i). For $v \in |N_u^- \cap C(u)|$, (u, v) is a disagreement. Right: Case 2a(ii). For $w \in |N_u^+ \cap C(u)|$ and $v \in N_w^+ \cap R_1'(u)$, (w, v) is a disagreement.

In total, we have

$$\boxed{S_{14}^+ \leqslant 2 \cdot 2^p \cdot \mathsf{OPT}^p = 2^{p+1} \cdot \mathsf{OPT}^p}$$

and $$\boxed{S_{12}^+ \leqslant 2^p \cdot S_{13}^+ + 8^p \cdot S_{14}^+ \leqslant 2^p \cdot 4^{p-1} \cdot ((20/3)^p + 2 + 2 \cdot 4^p) \cdot \mathsf{OPT}^p + 8^p \cdot \mathsf{OPT}^p.}$$

Next we turn to bounding S_{11}^+. Recall that $R_1 = \{u : |N_u^- \cap \{v : d_{uv} \leqslant 0.7\}| \geqslant \frac{10}{3} \cdot \Delta_u\}$ and

$$S_{11}^+ \leqslant \sum_{u \in R_1} |N_u^+ \cap C(u)|^p \leqslant \sum_{u \in R_1} |N_u^+|^p.$$

So it suffices to bound the right-hand side, which we do in the following proposition.

Proposition 3. Let R_1 be the set of u for which Step 3 of Definition 3 applies. Then

$$\sum_{u \in R_1} |N_u^+|^p \leqslant ((20/3)^p + 2 + 2 \cdot 4^p) \cdot \mathit{OPT}^p.$$

Proof of Proposition 3. For $u \in R_1$, define $R_1(u) = N_u^- \cap \{v : d_{uv} \leqslant 0.7\}$, so in particular $|R_1(u)| \geqslant \frac{10}{3} \cdot \Delta_u$. Fix a vertex $u \in R_1$. We consider a few cases. The crux of the argument is Case 2a(ii).

Case 1: At least a 0.15 fraction of N_u^+ is in clusters other than $C(u)$.
Let $u \in V^1$ be the vertices in this case. This means that $0.15 \cdot |N_u^+| \leqslant y(u)$, so

$$\boxed{\sum_{u \in V^1} |N_u^+|^p \leqslant \sum_{u \in V^1} \frac{1}{0.15^p} y(u)^p \leqslant (20/3)^p \cdot \mathsf{OPT}^p.}$$

Case 2: At least a 0.85 fraction of N_u^+ is in $C(u)$.
We further partition the cases based on how much $R_1(u)$ intersects $C(u)$.

Case 2a: At least half of $R_1(u)$ is in clusters other than $C(u)$.
We partition into cases (just one more time!) based on the size of $N_u^- \cap C(u)$. See Figure 2.

- Case 2a(i): At least half of $R_1(u)$ is in clusters other than $C(u)$ and $|N_u^- \cap C(u)| \geqslant \Delta_u$. Let $u \in V^{2a(i)}$ be the vertices in this case. Note that $y(u) \geqslant |N_u^- \cap C(u)|$. Then

$$\boxed{\sum_{u \in V^{2ai}} |N_u^+|^p = \sum_{u \in V^{2ai}} \Delta_u^p \leqslant \sum_{u \in V^{2ai}} |N_u^- \cap C(u)|^p \leqslant \sum_{u \in V^{2ai}} y(u)^p \leqslant \mathsf{OPT}^p.}$$

- Case 2a(ii): At least half of $R_1(u)$ is in clusters other than $C(u)$ and $|N_u^- \cap C(u)| \leqslant \Delta_u$.

  Let $u \in V^{2a(ii)}$ be the vertices in this case. Denote the vertices in $R_1(u)$ that are in clusters other than $C(u)$ by $R_1'(u)$. By definition of Case 2a(ii), $|R_1'(u)| \geqslant \frac{5}{3} \cdot \Delta_u$. A key fact we will use is that $|C(u)| \leqslant 2 \cdot \Delta_u$:

  $$|C(u)| = |N_u^- \cap C(u)| + |N_u^+ \cap C(u)| \leqslant \Delta_u + \Delta_u = 2 \cdot \Delta_u.$$

  For $u \in V^{2a(ii)}$ and $w \in N_u^+ \cap C(u)$, define $\varphi(u,w) = |R_1'(u) \cap N_w^+|$.

  Each $w \in N_u^+ \cap C(u)$ dispenses $\varphi(u,w)^p/|C(u)|$ charge to u. Also, observe that for $v \in R_1'(u)$, we have that $d_{uv} \leqslant 0.7$, so we know by Proposition 1 that $|N_u^+ \cap N_v^+ \cap C(u)| \geqslant 0.15 \cdot |N_u^+|$. This implies that

  $$\begin{aligned}\sum_{w \in N_u^+ \cap C(u)} |R_1'(u) \cap N_w^+| &= \sum_{w \in N_u^+ \cap C(u)} \sum_{v \in R_1'(u) \cap N_w^+} 1 = \sum_{v \in R_1'(u)} \sum_{\substack{w \in \cap N_v^+ \\ \cap C(u) \cap N_u^+}} 1 \\ &= \sum_{v \in R_1'(u)} |C(u) \cap N_u^+ \cap N_v^+| \geqslant \sum_{v \in R_1'(u)} 0.15 \cdot |N_u^+| \\ &= 0.15 \cdot |N_u^+| \cdot |R_1'(u)| \geqslant 0.15 \cdot \Delta_u \cdot \frac{5}{3}\Delta_u = 0.25 \cdot \Delta_u^2.\end{aligned}$$

  First we lower bound the amount of charge each u satisfying Case 2a(ii) receives, which is at least

  $$\begin{aligned}\frac{1}{|C(u)|} \sum_{w \in N_u^+ \cap C(u)} \varphi(u,w)^p &\geqslant \frac{1}{|C(u)|} \cdot \frac{1}{|N_u^+ \cap C(u)|^{p-1}} \cdot \Big(\sum_{w \in N_u^+ \cap C(u)} \varphi(u,w) \Big)^p \\ &\geqslant \frac{1}{2\Delta_u} \cdot \frac{1}{\Delta_u^{p-1}} \cdot \Big(0.25 \cdot \Delta_u^2\Big)^p \geqslant \frac{1}{2} \cdot 0.25^p \cdot |N_u^+|^p,\end{aligned}$$

  where in the first inequality we have applied Jensen's inequality.

  Next we need to upper bound the amount of charge dispensed in total to all u satisfying Case 2a(ii). Note by definition that $\varphi(u,w) \leqslant y(w)$. Each vertex $w \in V$ dispenses at most $y(w)^p/|C(u)| = y(w)^p/|C(w)|$ charge to each $u \in C(w) \cap N_w^+$. So in total each w dispenses at most $|C(w)| \cdot y(w)^p/|C(w)| = y(w)^p$ charge to all u satisfying Case 2a(ii).

  Now we put together the lower and upper bounds on the total charge dispensed:

  $$\sum_{w \in V} y(w)^p \geqslant \text{charge dispensed} \geqslant \sum_{u \in V^{2a(ii)}} \frac{1}{|C(u)|} \sum_{w \in N_u^+ \cap C(u)} \varphi(u,w)^p \geqslant \sum_{u \in V^{2a(ii)}} \frac{1}{2} \cdot 0.25^p \cdot |N_u^+|^p.$$

  In all, $\boxed{\sum_{u \in V^{2a(ii)}} |N_u^+|^p \leqslant 2 \cdot 4^p \cdot \sum_{w \in V} y(w)^p \leqslant 2 \cdot 4^p \cdot \mathsf{OPT}^p.}$

Case 2b: At least half of $R_1(u)$ is in $C(u)$.
Let $u \in V^{2b}$ be the vertices in this case. Denote the vertices in $R_1(u)$ that are in $C(u)$ by $R_1''(u)$. By definition of Case 2b, $|R_1''(u)| \geqslant \frac{5}{3} \cdot \Delta_u$. Since every vertex in $R''(u)$ is in N_u^-, there are at least $|R''(u)|$ disagreements incident to u. So $y(u) \geqslant |R''(u)| \geqslant \frac{5}{3} \cdot \Delta_u$, which gives that

$$\boxed{\sum_{u \in V^{2b}} |N_u^+|^p = \sum_{u \in V^{2b}} \Delta_u^p \leqslant \sum_{u \in V^{2b}} y(u)^p \leqslant \mathsf{OPT}^p.}$$

Adding the terms in the boxed expressions across all cases, the proposition follows. □

So we have $\boxed{S_{11}^+ \leqslant \sum_{u\in R_1} |N_u^+|^p \leqslant ((20/3)^p + 2 + 2\cdot 4^p)\cdot \mathsf{OPT}^p.}$

Adding together all the cases, we conclude that

$$(S^+)^p \leqslant 2^p\cdot(S_1^+ + S_2^+) \leqslant 2^p\cdot(S_{11}^+ S_{12^+} + S_2^+) \leqslant 2^p\cdot[(8^p/2+1)((20/3)^p+2+2\cdot 4^p)+8^p+1]\cdot \mathsf{OPT}^p.$$

□

4.2.2 Fractional cost of negative edges in ℓ_p-norms

This section bounds the cost of negative edges. The meanings of $\mathcal{C}$, $C(\cdot)$, and y are as in the previous subsection.

Lemma 6. For $p \in \mathbb{R}_{\geqslant 1}$, the fractional cost of the adjusted correlation metric f in the ℓ_p-norm objective for the set of negative edges is a constant factor approximation to the optimal, i.e.,

$$(S^-)^p = \sum_{u\in V}\Big(\sum_{v\in N_u^-}(1-f_{uv})\Big)^p \leqslant 2^p((200/9)^p + 1 + (10/3)^p + 2\cdot(20/3)^p)\cdot \mathit{OPT}^p.$$

Proof. We have

$$(S^-)^p = \sum_{u\in V}\Big(\sum_{v\in N_u^-}(1-f_{uv})\Big)^p \leqslant 2^p \underbrace{\sum_{u\in V}\Big(\sum_{v\in N_u^-\cap C(u)}(1-f_{uv})\Big)^p}_{S_1^-} + 2^p \underbrace{\sum_{u\in V}\Big(\sum_{v\in N_u^-\cap \overline{C(u)}}(1-f_{uv})\Big)^p}_{S_2^-}.$$

It is easy to bound S_1^- by using the trivial upper bound $1 - f_{uv} \leqslant 1$:

$$\boxed{S_1^- = \sum_{u\in V}\Big(\sum_{v\in N_u^-\cap C(u)}(1-f_{uv})\Big)^p \leqslant \sum_{u\in V}\Big(\sum_{v\in N_u^-\cap C(u)} 1\Big)^p \leqslant \sum_{u\in V} y(u)^p = \mathsf{OPT}^p,}$$

where we have used that every edge $(u, v) \in E^-$ with $v \in C(u)$ is a disagreement incident to u. Next, we bound S_2^-. Let R_1 and R_2 be as in the previous subsection: $R_1 = \{u : |N_u^- \cap \{v : d_{uv} \leqslant 0.7\}| \geqslant \frac{10}{3}\cdot\Delta_u\}$ and $R_2 = V\backslash R_1$. For $u \in R_2$, define

$$V_u = \{v : v \in N_u^- \cap \overline{C(u)}, d_{uv} \leqslant 0.7\}.$$

Note that the definition of V_u is the same as $R_1'(u)$ in the previous subsection, but here V_u is only defined for $u \in R_2$, while $R_1'(u)$ was defined for $u \in R_1$. For $u \in R_1$, we have $1 - f_{uv} = 0$ for every $v \in V\backslash\{u\}$. So the outer sum in S_2^- only need be taken over $u \in R_2$:

$$S_2^- = \sum_{u\in R_2}\Big(\sum_{v\in N_u^-\cap\overline{C(u)}}(1-f_{uv})\Big)^p \leqslant \sum_{u\in R_2}\Big(\sum_{\substack{v: v\in N_u^-\cap\overline{C(u)},\\ d_{uv}\leqslant 0.7}}(1-d_{uv})\Big)^p \leqslant \sum_{u\in R_2}|V_u|^p$$

In the second equality, we have used that if $u \in R_2$ and $v \in N_u^-$, then $f_{uv} = d_{uv}$, unless f_{uv} was rounded up to 1 in Step 2 of Definition 3 (which happens when $d_{uv} > 0.7$), or f_{uv} was rounded up to 1 in Step 3 (in which case $1 - f_{uv} = 0 \leqslant 1 - d_{uv}$).

A key observation is that since $u \in R_2$, it is the case that $|V_u| \leqslant \frac{10}{3}\cdot\Delta_u$.

Fix a vertex $u \in R_2$. We consider a few cases.

Case 1: At least a 0.15 fraction of N_u^+ is in clusters other than $C(u)$.
Define V^1 to be the set of $u \in R_2$ that satisfy Case 1. Then for $u \in V^1$, $0.15 \cdot |N_u^+| \leqslant y(u)$, and further

$$|V_u| \leqslant \frac{10}{3}\Delta_u \leqslant \frac{1}{0.15} \cdot \frac{10}{3} y(u) = \frac{200}{9} y(u).$$

It follows that

$$\boxed{\sum_{u \in V^1} |V_u|^p \leqslant (200/9)^p \cdot \sum_{u \in V^1} y(u)^p \leqslant (200/9)^p \cdot \mathsf{OPT}^p.}$$

Case 2: At least a 0.85 fraction of N_u^+ is in $C(u)$.
Define V^2 to be the set of $u \in R_2$ that satisfy Case 2. Fix $u \in V^2$ and $v \in V_u$. Define $N_{u,v} = N_u^+ \cap N_v^+ \cap C(u)$. Since $d_{uv} \leqslant 0.7$ and by the assumption of this case, using Proposition 1 we have

$$|N_{u,v}| = |N_u^+ \cap N_v^+ \cap C(u)| \geqslant 0.15 \cdot \Delta_u.$$

Observe that since $v \notin C(u)$ for $v \in V_u$, (v, w) is a (positive) disagreement for all $w \in N_{u,v}$.
Case 2a: $|N_u^- \cap C(u)| \geqslant \Delta_u$.

Define V^{2a} to be the set of $u \in V^2$ that satisfy Case 2a. Since all edges (u, v) with $v \in N_u^- \cap C(u)$ are disagreements, we have $y(u) \geqslant \Delta_u$. Recalling that $|V_u| \leqslant \frac{10}{3} \cdot \Delta_u$ for $u \in R_2$, we have

$$\boxed{\sum_{u \in V^{2a}} |V_u|^p \leqslant \sum_{u \in V^{2a}} (10/3 \cdot \Delta_u)^p \leqslant (10/3)^p \cdot \sum_{u \in V^{2a}} y(u)^p \leqslant (10/3)^p \cdot \mathsf{OPT}^p.}$$

Case 2b: $|C(u)| \leqslant 2\Delta_u$.

Define V^{2b} to be the $u \in V^2$ satisfying Case 2b. Fix $w \in N_u^+ \cap C(u)$ and $u \in V^{2b}$. Define

$$\varphi(u, w) = |V_u \cap N_w^+|,$$

i.e. $\varphi(u, w)$ is the number of $v \in V_u$ with $w \in N_{u,v}$. Each $w \in N_u^+ \cap C(u)$ dispenses $\frac{\varphi(u,w)^p}{|C(u)|}$ charge to u. Also,

$$\begin{aligned}\sum_{w \in N_u^+ \cap C(u)} \varphi(u, w) &= \sum_{w \in N_u^+ \cap C(u)} |V_u \cap N_w^+| = \sum_{w \in N_u^+ \cap C(u)} \sum_{v \in V_u \cap N_w^+} 1 \\ &= \sum_{v \in V_u} \sum_{w \in N_{u,v}} 1 = \sum_{v \in V_u} |N_{u,v}| \geqslant |V_u| \cdot 0.15 \cdot \Delta_u.\end{aligned}$$

Now we can lower bound the amount of charge each u satisfying Case 2b receives, which is at least

$$\begin{aligned}\frac{1}{|C(u)|} \sum_{w \in N_u^+ \cap C(u)} \varphi(u, w)^p &\geqslant \frac{1}{|C(u)|} \cdot \frac{1}{|N_u^+ \cap C(u)|^{p-1}} \Big(\sum_{w \in N_u^+ \cap C(u)} \varphi(u, w) \Big)^p \\ &\geqslant \frac{1}{2\Delta_u} \cdot \frac{1}{\Delta_u^{p-1}} \left(|V_u| \cdot 0.15 \cdot \Delta_u \right)^p = \frac{1}{2} \cdot 0.15^p \cdot |V_u|^p,\end{aligned}$$

where in the first line we used Jensen's inequality.

To upper bound the amount of charge dispensed in total to all u satisfying Case 2b, first note that $\varphi(u, w) \leqslant y(w)$. Also, each vertex $w \in V$ only distributes charge to $u \in C(w) \cap N_w^+$, and the amount of charge distributed to each such u is

$$\frac{\varphi(u, w)^p}{|C(u)|} = \frac{\varphi(u, w)^p}{|C(w)|} \leqslant \frac{y(w)^p}{|C(w)|},$$

so that in total each w dispenses at most $\frac{y(w)^p}{|C(w)|} \cdot |C(w)| \leqslant y(w)^p$ charge. Putting together the lower and upper bounds on the amount of charge dispensed:

$$\sum_{w \in V} y(w)^p \geqslant \text{total charge dispensed} \geqslant \sum_{u \in V^{2b}} \frac{1}{|C(u)|} \sum_{w \in N_u^+ \cap C(u)} \varphi(u,w)^p \geqslant \sum_{u \in V^{2b}} \frac{1}{2} \cdot 0.15^p \cdot |V_u|^p.$$

So in all,

$$\boxed{\sum_{u \in V^{2b}} |V_u|^p \leqslant 2 \cdot (20/3)^p \cdot \sum_{w \in V} y(w)^p = 2 \cdot (20/3)^p \cdot \mathsf{OPT}^p.}$$

Adding together all the cases, we see that $(S^-)^p \leqslant 2^p((200/9)^p + 1 + (10/3)^p + 2 \cdot (20/3)^p) \cdot \mathsf{OPT}^p$. □

4.3 Proofs of Theorem 1 and Corollary 1

Here we show that Theorem 1 and Corollary 1 follow directly from the preceding lemmas.

Proof of Theorem 1. First we show that Lemma 4 implies that the clustering resulting from inputting f into the KMZ rounding algorithm is $O(1)$-approximate in any ℓ_p-norm. Since the rounding algorithm does not depend on p, the clustering will be the same for all p. Let $\mathcal{C}^*$ be the clustering produced by running the KMZ rounding algorithm with the adjusted correlation metric f as input. Let $\mathsf{ALG}(u)$ be the number of edges incident to u that are disagreements with respect to $\mathcal{C}^*$. From [28] and Lemmas 1 and 2, we have that for every $u \in V$, $\mathsf{ALG}(u) \leqslant 12 \cdot y_u$ where y_u is as in LP 2.2 when taking $x = f$. So $||y||_p$ is the fractional cost of f in the ℓ_p-norm. In what follows, $||\mathsf{ALG}||_p$ is the objective value of $\mathcal{C}^*$ in the ℓ_p-norm and $\mathsf{OPT}(p)$ is the optimal objective value in the ℓ_p-norm. Thus using Lemma 4 in the last inequality, we have

$$||\mathsf{ALG}||_p \leqslant 12 \cdot ||y||_p \leqslant 12 \cdot 529 \cdot \mathsf{OPT}(p) = 6348 \cdot \mathsf{OPT}(p).$$

We show the overall run-time is $O(n^\omega)$. Recall from the analysis in [19] that computing the correlation metric takes time $O(n^\omega)$, and the KMZ rounding algorithm takes time $O(n^2)$. We just have to show the post-processing of d in Steps 2 and 3 of Definition 3 that were done in order to obtain the adjusted correlation metric f can be done quickly. Indeed, Step 2 takes $O(n^2)$ time as it simply iterates through the edges. Step 3 also takes $O(n^2)$ time, since it visits each vertex and iterates through the neighbors. Thus, the run-time remains $O(n^\omega)$. □

Proof of Corollary 1. In [19], they observe that to reduce the run-time from $O(n^\omega)$ to $O(n\Delta^2 \log n)$ for graphs with maximum positive degree bounded by Δ, one can compute the correlation metric d in $O(n\Delta^2)$ time and then run the KMZ rounding algorithm in $O(n\Delta^2 \log n)$ time (see the proof of Corollary 1.2 in Appendix D of [19]). We need only compute f_{uv} when $f_{uv} < 1$. Otherwise, we can handle $f_{uv} = 1$ implicitly. As in [19], for each $u \in V$, we can maintain a list of d_{uv} for all v with $d_{uv} < 1$. Computing these lists takes $O(n\Delta^2)$ time in total. (This is because there are at most Δ^2 vertices v that are distance two away from u in the positive subgraph.) Steps 2 and 3 only prune these lists further, since some distances that are below 1 may be raised to 1; importantly, no distance that is already equal to 1 under d will be reduced in Steps 2 and 3. To carry out Step 2, for each vertex u it takes $O(\Delta^2)$ time to iterate through the list for u and raise the appropriate d_{uv} to 1. Similarly, to carry out Step 3, for each vertex u it takes $O(\Delta^2)$ time to determine whether the condition in Step 3 is satisfied; if it is, we just handle the vertex u implicitly, as all distances f_{uv} are set to 1.

Since each vertex's list still has size at most Δ^2 after the post-processing in Steps 2 and 3, the KMZ rounding algorithm with f as input takes time $O(n\Delta^2 \log n)$, by the exact same argument as in [19]. □

5 Conclusion

This paper considered correlation clustering on unweighted, complete graphs, a problem that arises in many settings including community detection and the study of large networks. All previous works that study minimizing the ℓ_p-norm (for $p \in \mathbb{R}_{>1}$) of the disagreement vector rely on solving a large, convex relaxation (which is costly to the algorithm's run-time) and produce a solution that is only $O(1)$-approximate for one specific value of p. We innovate upon this rich line of work by (1) giving the first combinatorial algorithm for the ℓ_p-norms for $p \in \mathbb{R}_{>1}$, (2) designing scalable algorithms for this practical problem, and (3) obtaining solutions that are $O(1)$-approximate for all ℓ_p-norms (for $p \in \mathbb{R}_{\geqslant 1} \cup \{\infty\}$) simultaneously. We note this last point is particularly important, as such solutions are good in both global and local senses, and thus may be more desirable than typical optimal or approximate solutions for correlation clustering. The existence of these solutions reveals a surprising structural property of correlation clustering.

One question is whether there is a simpler existential (not necessarily algorithmic) proof that there exists an $O(1)$-approximation for the all-norm objective for correlation clustering.

It is also of interest to implement the KMZ algorithm with the adjusted correlation metric as input, and empirically gain an understanding of how good the adjusted correlation metric is for different ℓ_p-norms. We suspect that our analysis is lossy (for instance, we did not attempt to optimize constants), and that the approximation obtained would be of much better quality than our analysis guarantees.

Finally, it would be interesting if an analogous result can be obtained for weighted correlation clustering.

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A Triangle Inequality

Below is the proof that the adjusted correlation metric f satisfies an approximate triangle inequality.

Proof of Lemma 1. Note that after Step 1 of Definition 3, f satisfies the triangle inequality, since $f = d$ here and for any u, v, w, $d_{uv} \leqslant d_{uw} + d_{vw}$. We then consider the execution of Step 2 in Definition 3. The triangle inequality on the adjusted correlation metric holds if either of (u, w) or (w, v) have distance 1 with respect to the adjusted correlation metric, and it also clearly holds if the three edges have the same distance as in the correlation metric. It follows that the only case to check is when $f_{uv} = 1$, $f_{uw} = d_{uw}$, and $f_{wv} = d_{wv}$. If $f_{uv} = 1$ at the end of Step 2, by the definition of f it must be that $d_{uv} > 0.7$. Since $d_{uv} \leqslant d_{uw} + d_{vw}$, we have that $0.7 \leqslant d_{uw} + d_{vw}$, and so $f_{uv} = 1 \leqslant \frac{10}{7}(d_{uw} + d_{vw}) \leqslant \frac{10}{7}(f_{uw} + f_{vw})$ at the end of Step 2.

Finally, Step 3 of Definition 3 also preserves this approximate triangle inequality. The f values on the edges leaving a vertex are raised to 1 together, and any triangle involving one of these edges necessarily involves another. □

B ℓ_∞-norm Approximation

Below we include the proof that the adjusted correlation metric f has bounded fractional cost in the ℓ_∞-norm.

Proof of Lemma 3. The fractional cost of the correlation metric d is at most $8 \cdot \mathsf{OPT}$ by [19]. That is,

$$\sum_{v \in N_u^+} d_{uv} + \sum_{v \in N_u^-} (1 - d_{uv}) \leqslant 8 \cdot \mathsf{OPT} \quad \text{for all } u \in V. \tag{4}$$

Consider any $u \in V$. Observe that the second sum in (4) can only decrease if Step 2 in Definition 3 is executed for any of the negative edges incident to u. Now let R_1 denote the set of vertices u such that Step 3 is executed on u. For $u \in R_1$, the second sum decreases by at least $\frac{3}{10} \cdot \frac{10}{3} \cdot \Delta_u = \Delta_u$, since there are at least $\frac{10}{3} \cdot \Delta_u$ negative edges (u, v) with $1 - d_{uv} \geqslant \frac{3}{10}$, and these edges are all rounded up to 1 in f, so $1 - d_{uv}$ is rounded down to 0. On the other hand, the first sum increases by at most Δ_u, since there are Δ_u positive edges incident to u and each gets rounded up to 1 in f. Therefore, we know that (4) does not increase when we replace d with f when $u \in R_1$.

Finally, it is possible that (4) increases even when $u \notin R_1$. This is because u may have neighbors v such that $v \in R_1$. For such v, d_{uv} is raised to $f_{uv} = 1$. This is only better for the second sum in (4). Now we consider what happens to the first sum in (4) when we replace d with f. Let $R_2 = V \backslash R_1$. Fix $u \in R_2$. Then

$$\begin{aligned}
\sum_{v \in N_u^+} f_{uv} &= \sum_{v \in N_u^+ \cap R_1} 1 + \sum_{v \in N_u^+ \cap R_2} d_{uv} \\
&= \sum_{\substack{v \in N_u^+ \cap R_1 \\ d_{uv} \leqslant 1/4}} 1 + \sum_{\substack{v \in N_u^+ \cap R_1 \\ d_{uv} > 1/4}} 1 + \sum_{v \in N_u^+ \cap R_2} d_{uv} \\
&\leqslant \sum_{\substack{v \in N_u^+ \cap R_1 \\ d_{uv} \leqslant 1/4}} 1 + \sum_{\substack{v \in N_u^+ \cap R_1 \\ d_{uv} > 1/4}} 4 \cdot d_{uv} + \sum_{v \in N_u^+ \cap R_2} d_{uv} \\
&\leqslant \sum_{\substack{v \in N_u^+ \cap R_1 \\ d_{uv} \leqslant 1/4}} 1 + 4 \cdot \sum_{v \in N_u^+} d_{uv}
\end{aligned}$$

The second term above is bounded by $32 \cdot \mathsf{OPT}$ by (4). To bound the first term, let w^* be an arbitrary vertex in $N_u^+ \cap R_1$ (if this set is empty, we just ignore the first term). Since $d_{uw^*} \leqslant 1/4$, we know by Proposition 2 that $\Delta_u \leqslant \frac{7}{3} \cdot \Delta_{w^*}$. Also,

$$\sum_{v \in N_{w^*}^+} f_{w^*v} + \sum_{v \in N_{w^*}^-} (1 - f_{w^*v}) \leqslant \Delta_w^*$$

since $w^* \in R_1$, and by the argument in the previous paragraph we have that $\Delta_{w^*} \leqslant 8 \cdot \mathsf{OPT}$. So

$$\sum_{\substack{v \in N_u^+ \cap R_1 \\ d_{uv} \leqslant 1/4}} 1 \leqslant \Delta_u \leqslant \frac{7}{3} \cdot \Delta_{w^*} \leqslant \frac{7}{3} \cdot 8 \cdot \mathsf{OPT}.$$

This concludes the proof. □

C ℓ_1-norm Approximation

The next lemma proves that the adjusted correlation metric well approximates the optimal in the ℓ_1-norm. We will show that the fractional cost of the adjusted correlation metric f in the ℓ_1-norm is a constant factor approximation to the optimal, i.e.,

$$\sum_{u \in V} \sum_{v \in N_u^+} f_{uv} + \sum_{u \in V} \sum_{v \in N_u^-} (1 - f_{uv}) = O(\mathsf{OPT}).$$

We argue the second sum—over the negative edges—has a constant factor approximation in the first lemma (Lemma 9), then show the first sum—over the positive edges—does too in the second lemma (Lemma 8). Combining these lemmas gives the following approximation guarantee for the ℓ_1-norm.

Lemma 7. The fractional cost of the adjusted correlation metric f in the ℓ_1-norm objective is a constant factor away from the cost of the optimal integral solution in the ℓ_1-norm

Proof. The fractional cost in the ℓ_1-norm is

$$\sum_{u \in V} \sum_{v \in N_u^+} f_{uv} + \sum_{u \in V} \sum_{v \in N_u^-} (1 - f_{uv}) \leqslant 74 \cdot \mathsf{OPT}.$$

The first sum is bounded by $34 \cdot \mathsf{OPT}$ by Lemma 8 and the second sum is bounded by $40 \cdot \mathsf{OPT}$ by Lemma 9. □

Lemma 8. The fractional cost of the adjusted correlation metric f in the ℓ_1-norm objective for the set of positive edges is a constant factor approximation to the optimal, i.e.,

$$\sum_{u \in V} \sum_{v \in N_u^+} f_{uv} = 34 \cdot \mathit{OPT}.$$

Proof of Lemma 8. Fix an optimal clustering $\mathcal{C}$. We partition vertices based on membership in $C(u)$ or $\overline{C(u)}$ and let y denote the disagreement vector of $\mathcal{C}$. We prove in the following claim that the total fractional cost of edges $e \in E^+$ such that $f_e = d_e$ is bounded. Then, it just remains to bound the fractional cost of positive edges whose distances are raised to 1 in Step 3 of Definition 3.

Claim 1. The total fractional cost of positive edges for the ℓ_1-norm objective when $f_e = d_e$ is a 3-approximation.

Proof of Claim 1. For the ℓ_1-norm, $\mathsf{OPT} = \sum_{u \in V} y(u)$. So for a fixed vertex u, we partition the fractional cost of disagreements from positive edges incident to u, i.e. $(u, v) \in E^+$, based on whether vertices v are in $C(u)$ or not:

$$\begin{aligned}\sum_{u \in V} \sum_{v \in N_u^+} f_{uv} &= \sum_{u \in V} \sum_{v \in N_u^+ \cap C(u)} d_{uv} + \sum_{u \in V} \sum_{v \in N_u^+ \cap \overline{C(u)}} d_{uv} \\ &= \sum_{u \in V} \sum_{v \in N_u^+ \cap C(u)} \frac{|N_u^+ \cap N_v^-| + |N_u^- \cap N_v^+|}{n - |N_u^- \cap N_v^-|} + \sum_{u \in V} \sum_{v \in N_u^+ \cap \overline{C(u)}} d_{uv}.\end{aligned}$$

The right-hand sum is easy as we can just use that d_{uv} is at most 1.

$$\boxed{\sum_{u \in V} \sum_{v \in N_u^+ \cap \overline{C(u)}} d_{uv} \leqslant \sum_{u \in V} \sum_{v \in N_u^+ \cap \overline{C(u)}} 1 \leqslant \sum_{u \in V} y(u) = \mathsf{OPT}}$$

For the left-hand sum, we will exploit that d_{uv} is symmetric, and specifically that we can bound the denominator by $|N_u^+|$ or $|N_v^+|$.

$$\begin{aligned}&\sum_{u \in V} \sum_{v \in N_u^+ \cap C(u)} \frac{|N_u^+ \cap N_v^-| + |N_u^- \cap N_v^+|}{n - |N_u^- \cap N_v^-|} \leqslant \sum_{u \in V} \sum_{v \in N_u^+ \cap C(u)} \frac{y(u) + y(v)}{n - |N_u^- \cap N_v^-|} \\ &\leqslant \sum_{u \in V} \sum_{v \in N_u^+} \frac{y(u) + y(v)}{n - |N_u^- \cap N_v^-|} = \sum_{u \in V} \sum_{v \in N_u^+} \frac{y(u)}{n - |N_u^- \cap N_v^-|} + \sum_{u \in V} \sum_{v \in N_u^+} \frac{y(v)}{n - |N_u^- \cap N_v^-|} \\ &= \sum_{u \in V} \sum_{v \in N_u^+} \frac{y(u)}{n - |N_u^- \cap N_v^-|} + \sum_{v \in V} \sum_{u \in N_v^+} \frac{y(v)}{n - |N_u^- \cap N_v^-|} \\ &\leqslant \sum_{u \in V} \frac{y(u)}{|N_u^+|} \cdot |N_u^+| + \sum_{v \in V} \frac{y(v)}{|N_v^+|} \cdot |N_v^+| \leqslant 2 \cdot \mathsf{OPT}.\end{aligned}$$

All in all:

$$\boxed{\sum_{u \in V} \sum_{v \in N_u^+ \cap C(u)} \frac{|N_u^+ \cap N_v^-| + |N_u^- \cap N_v^+|}{n - |N_u^- \cap N_v^-|} \leqslant 2 \cdot \mathsf{OPT}}$$

□

Now, we must bound the cost of the positive edges whose distances were raised in Step 3 of Definition 3. Let $R_1 = \{u : |\{N_u^- \cap \{v : d_{uv} \leqslant 0.7\}| \geqslant \frac{10}{3} \cdot \Delta_u\}$. For any vertex u, define the shorthand $R_1(u) = \{N_u^- \cap \{v : d_{uv} \leqslant 0.7\}\}$ [11]; recall that $u \in R_1$ is put in its own cluster by the adjusted correlation metric. Then, our next goal is to show that

$$2 \cdot \sum_{u \in R_1} |N_u^+| = O(\mathsf{OPT}). \tag{5}$$

Note that the factor of 2 comes from the observation that even if $u \notin R_1$, u may have neighbors $v \in R_1 \cap N_u^+$, in which cases f_{uv} is raised to 1 even though Step 3 is not executed on u. Fix a $u \in R_1$. We subdivide into several cases based on $\mathcal{C}$.

[11] The use of a subscript may not be clear here, but it is helpful in the ℓ_p-norm proof.

Case 1: At least a 0.15 fraction of N_u^+ is in clusters other than $C(u)$.
Let $u \in V^1$ be the vertices in this case. There are at least $0.15 \cdot |N_u^+|$ disagreements incident to u, so

$$\boxed{\sum_{u \in V^1} |N_u^+| \leqslant \sum_{u \in V^1} \frac{1}{0.15} \cdot y(u) \leqslant \frac{20}{3} \cdot \mathsf{OPT}.}$$

Case 2: At least a 0.85 fraction of N_u^+ is in $C(u)$.
We subdivide Case 2 into two cases depending on how much of $R_1(u)$ is in $\overline{C(u)}$.

Case 2a: At least half of $R_1(u)$ is in clusters other than $C(u)$.
Let $u \in V^{2a}$ be the vertices in this case. Denote this subset of $R_1(u)$ by $R_1'(u) = R_1(u) \cap \overline{C(u)}$, and note that $|R_1'(u)| \geqslant \frac{5}{3} \cdot \Delta_u$. If the number of negative edges incident to u that are contained in $C(u)$ is more than Δ_u, then charge $|N_u^+| = \Delta_u$ to $y(u)$, since these negative edges are disagreements. These $u \in V^{2a}$ thus contribute OPT to (5).

Otherwise, $|C(u)| \leqslant 2\Delta_u$. Consider $v \in R_1'(u)$. Again, let $N_{u,v} = N_u^+ \cap N_v^+ \cap C(u)$. Since $d_{uv} \leqslant 0.7$ and by the assumption bringing us into Case 2, we apply Proposition 1 to see that $|N_{u,v}| \geqslant 0.15 \cdot |N_u^+|$. For each $v \in R_1'(u)$ and $w \in N_{u,v}$, the positive edge (w, v) is a disagreement.

For $u \in V^{2a}$ and $w \in N_u^+ \cap C(u)$, define

$$\varphi(u, w) = |R_1'(u) \cap N_w^+|.$$

Let each $w \in N_u^+ \cap C(u)$ dispense charge $\varphi(u, w)/|C(u)|$ to u. First, observe that u receives $\Omega(|N_u^+|)$ charge; specifically, u receives charge at least

$$\begin{aligned}
\sum_{w \in N_u^+ \cap C(u)} \frac{\varphi(u,w)}{|C(u)|} &= \sum_{w \in N_u^+ \cap C(u)} \frac{|R_1'(u) \cap N_w^+|}{|C(u)|} \\
&= \frac{1}{|C(u)|} \cdot \sum_{w \in N_u^+ \cap C(u)} \sum_{v \in R_1'(u) \cap N_w^+} 1 = \frac{1}{|C(u)|} \cdot \sum_{v \in R_1'(u)} \sum_{\substack{w \in \cap N_v^+ \\ \cap C(u) \cap N_u^+}} 1 \\
&= \frac{1}{|C(u)|} \cdot \sum_{v \in R_1'(u)} |C(u) \cap N_u^+ \cap N_v^+| \geqslant \frac{1}{|C(u)|} \cdot \sum_{v \in R_1'(u)} 0.15 \cdot |N_u^+| \\
&\geqslant \frac{1}{|C(u)|} \cdot 0.15 \cdot |N_u^+| \cdot |R_1'(u)| \geqslant \frac{1}{2\Delta_u} \cdot 0.15 \cdot \Delta_u \cdot \frac{5}{3}\Delta_u = \frac{1}{8} \cdot \Delta_u.
\end{aligned}$$

Next we upper bound the charge dispensed in total to all $u \in V^{2a}$. Note by definition that $\varphi(u, w) \leqslant y(w)$. Every vertex $w \in V$ dispenses at most

$$\frac{y(w)}{|C(u)|} = \frac{y(w)}{|C(w)|}$$

charge to each $u \in C(w) \cap N_w^+$. So each w dispenses a total of charge at most

$$\frac{y(w)}{|C(w)|} \cdot |C(w)| = y(w)$$

to all $u \in V^{2a}$. Putting the lower and upper bounds together:

$$\sum_{w \in V} y(w) \geqslant \text{total charge dispensed}$$

$$\geqslant \sum_{u\in V^{2a}:|C(u)|\leqslant 2\Delta_u} \frac{1}{|C(u)|} \sum_{w\in N_u^+\cap C(u)} \varphi(u,w)$$
$$\geqslant \sum_{u\in V^{2a}:|C(u)|\leqslant 2\Delta_u} \frac{1}{8}\cdot |N_u^+|$$

so

$$\boxed{\sum_{u\in V^{2a}} |N_u^+| \leqslant \mathsf{OPT} + 8\cdot \sum_{w\in V} y(w) \leqslant 9\cdot \mathsf{OPT}.}$$

Case 2b: At least half of $R_1(u)$ is in $C(u)$.
Let $u \in V^{2b}$ be the vertices in this case. There are at least $\frac{5}{3}\Delta_u$ negative disagreeing edges incident to u (those edges (u,w) with $w \in R_1(u)$). So we may charge $|N_u^+|$ to $y(u)$:

$$\boxed{\sum_{u\in V^{2b}} |N_u^+| \leqslant \mathsf{OPT}}$$

This concludes the bounding of the fractional cost of the negative edges, and our constant in the approximation factor is $3 + 20/3 + 9 + 1 < 20$ (adding together the boxed upper bounds). □

Lemma 9. The fractional cost of the adjusted correlation metric f in the ℓ_1-norm objective for the set of negative edges is a constant factor approximation to the optimal, i.e.,

$$\sum_{u\in V}\sum_{v\in N_u^-} (1 - f_{uv}) \leqslant 40\cdot \mathit{OPT}.$$

Proof of Lemma 9. First observe that negative edges that have been rounded up to 1 contribute 0 to the fractional cost. So we may focus on the fractional cost of negative edges $e = (u,v)$ with $d_{uv} = f_{uv} \leqslant 0.7$.

Fix an optimal clustering $\mathcal{C}$, whose value is OPT. If $d_{uv} \leqslant 0.7$ but u and v are in the same cluster in $\mathcal{C}$, then (u,v) is a disagreement with respect to $\mathcal{C}$, so we can charge the fractional cost $1 - f_e = 1 - d_e \leqslant 1$ to $\boxed{\mathsf{OPT}}$.

Now we need to consider the negative edges whose endpoints are in different clusters in $\mathcal{C}$. Recall $C(u)$ is the cluster of $\mathcal{C}$ containing vertex u. We define the set of cut edges incident to u to be

$$E_u = \{(u,v) : (u,v) \in E^-, d_{uv} \leqslant 0.7, C(u) \neq C(v)\}.$$

By Step 3 in the definition of the adjusted correlation metric, $|E_u| \leqslant 10/3\cdot\Delta_u$. Since the fractional cost of each cut edge is $1 - f_{uv} = 1 - d_{uv} \geqslant 0.3 = \Omega(1)$, we need to show that the total number of cut edges is bounded by OPT, i.e., $\sum_{u\in V}|E_u| = O(\mathsf{OPT})$. We let $V_u = \{v : (u,v) \in E_u\}$, so our goal is equivalent to showing that $\sum_{u\in V}|V_u| = O(\mathsf{OPT})$. The definition of V_u is the same as $R_1'(u)$ in the proof of the previous lemma, but $R_1'(u)$ was only defined for $u \in R_1$.

We will charge the value of $\sum_{u\in V}|V_u|$ to disagreements in $\mathcal{C}$. We use $y(u)$ to denote the number of disagreements incident to u with respect to $\mathcal{C}$.

Fix a vertex u. We consider several cases based on the clustering $\mathcal{C}$.

Case 1: At least a 0.15 fraction of N_u^+ is in clusters other than $C(u)$.
Let V^1 be the vertices of V in this case. For any such $u \in V^1$, there are at least $0.15 \cdot \Delta_u$ disagreements incident to u in $\mathcal{C}$, so $0.15 \cdot \Delta_u \leqslant y(u)$. Combining this with the fact that $|V_u| \leqslant \frac{10}{3} \cdot \Delta_u$, we see that

$$|V_u| \leqslant \frac{10}{3} \cdot \Delta_u \leqslant \frac{200}{9} \cdot y(u).$$

Summing over all $u \in V^1$, we see that

$$\boxed{\sum_{u \in V^1} |V_u| \leqslant \frac{200}{9} \cdot \sum_{u \in V} y(u) = \frac{200}{9} \cdot \mathsf{OPT}.}$$

Case 2: At least a 0.85 fraction of N_u^+ is in $C(u)$.
Let V^2 be the vertices of V in this case. For any such $u \in V^2$, we will show that for every $v \in V_u$, there is large overlap between $C(u)$, N_u^+, and N_v^+. Proposition 1 will help us as we further refine Case 2 into two settings:

Case 2a: At least 0.85 fraction of N_u^+ is in $C(u)$, and there are at least Δ_u negative edges incident to u with both endpoints in $C(u)$.
Let V^{2a} be the vertices of V^2 in this case. Since these negative edges are disagreements, we may charge $|V_u|$ to $y(u)$:

$$\boxed{\sum_{u \in V^{2a}} |V_u| \leqslant \frac{10}{3} \cdot \sum_{u \in V} \Delta_u \leqslant \frac{10}{3} \cdot \sum_u y(u) = \frac{10}{3} \cdot \mathsf{OPT}.}$$

Case 2b: At least 0.85 fraction of N_u^+ is in $C(u)$, and $|C(u)| \leqslant 2 \cdot \Delta_u$.
Observe Cases 2a and 2b cover the full range of Case 2. This is because we can write $|C(u)| = |C(u) \cap N_u^+| + |C(u) \cap N_u^-|$, and if there are at most Δ_u negative edges incident to u with both endpoints in $C(u)$, this is equivalent to saying $|C(u) \cap N_u^-| \leqslant \Delta_u$. So, both $|C(u) \cap N_u^+|$ and $|C(u) \cap N_u^-|$ are upper bounded by Δ_u (since $\Delta_u = |N_u^+|$).

Let V^{2b} be the vertices of V^2 in this case. Let $N_{u,v} := N_v^+ \cap N_u^+ \cap C(u)$. We know from Proposition 1 (for $v \in R_1(u)'$ and using the assumption bringing us into Case 2) that $|N_{u,v}| \geqslant 0.15 \cdot \Delta_u$. Observe that for $v \in V_u \subseteq N_u^- \cap \overline{C(u)}$ and $w \in N_{u,v}$, the positive edge (w, v) is a disagreement.

Fix $w \in N_u^+ \cap C(u)$ and $u \in V^{2b}$. Define $\varphi(u, w) = |V_u \cap N_w^+|$. We say that w dispenses $\varphi(u, w)/|C(u)|$ charge to u. We would like to show that each $u \in V^{2b}$ receives $\Omega(|V_u|)$ charge, and that each w dispenses at most $y(w)$ charge. By double counting,

$$\sum_{w \in N_u^+ \cap C(u)} \varphi(u, w) \geqslant |V_u| \cdot 0.15\Delta_u$$

where we have used that $\varphi(u, w)$ is equal to the number of $v \in V_u$ such that $w \in N_{u,v}$, and we know that $|N_{u,v}|$ is at least $0.15\Delta_u$. So, the total amount of charge that $u \in V^{2b}$ receives is

$$\sum_{w \in N_u^+ \cap C(u)} \frac{\varphi(u, w)}{|C(u)|} \geqslant |V_u| \cdot 0.15\Delta_u/|C(u)| \geqslant 3/40 \cdot |V_u|$$

since $|C(u)| \leqslant 2\Delta_u$ by assumption of the case.

Now we need to upper bound the amount of charge that each $w \in V$ dispenses in total. Note that $\varphi(u,w) \leqslant y(w)$, and w only dispenses to vertices $u \in C(w) \cap N_w^+$. Further, $C(u) = C(w)$. So w dispenses at most $|C(w)| \cdot y(w)/|C(u)| = y(w)$ charge in total. Putting everything together:

$$\sum_{w \in V} y(w) \geqslant \text{total charge dispensed}$$
$$\geqslant \sum_{u \in V^{2b}} \sum_{w \in N_u^+ \cap C(u)} \frac{\varphi(u,w)}{|C(u)|} \geqslant \sum_{u \in V^{2b}} \frac{3}{40} \cdot |V_u|.$$

So in all,

$$\boxed{\sum_{u \in V^{2b}} |V_u| \leqslant \frac{40}{3} \cdot \sum_{w \in V} y(w) = \frac{40}{3} \cdot \mathsf{OPT}.}$$

This concludes the bounding of the fractional cost of the negative edges, and our constant in the approximation factor is $1 + 200/9 + 10/3 + 40/3 < 40$ (adding together the boxed upper bounds). □

D KMZ Rounding Algorithm and Run-time

We construct a fractional solution with our adjusted correlation metric (recall we use this in place of an LP solution), and then input this solution to a rounding algorithm by Kalhan, Makarychev, and Zhou [2019].

First, we set up some notation for the rounding algorithm, mainly following the example set by Kalhan, Makarychev, and Zhou. Let the ball of radius ρ, with respect to the distance defined by any semi-metric z on V, around vertex $u \in V$ be denoted $\mathsf{Ball}(u,\rho) = \{v \in V \mid z_{uv} \leqslant \rho\}$. The algorithm is iterative, and continues to find new clusters until every vertex is clustered. Let the set of unclustered vertices at time t be denoted by $V_t \subseteq V$. The cluster center in V_t is the vertex maximizing the quantity

$$L_t(u) = \sum_{v \in \mathsf{Ball}(u,r) \cap V_t} (r - z_{uv})$$

for r a parameter input to the algorithm. Note that when L_t is large, vertices in $\mathsf{Ball}(u,r) \cap V_t$ are closely clustered together.

Algorithm 1. [Rounding algorithm]

Input: Semi-metric z on V.
Output: Clustering $\mathcal{C}$.

1. Let $V_0 = V$, $r = 1/5$, $t = 0$.
2. while $(V_t \neq \varnothing)$
   - Find $u_t^* = \arg\max_{u \in V_t} L_t(u) = \arg\max_{u \in V_t} \sum_{v \in \mathrm{Ball}(u,r) \cap V_t} r - z_{uv}$.
   - Create $C_t = \mathrm{Ball}(u_t^*, 2r) \cap V_t$.
   - Set $V_{t+1} = V_t \backslash C_t$ and $t = t + 1$.
3. Return $\mathcal{C} = (C_0, \ldots, C_{t-1})$.

We define $\mathsf{LP}(u, v)$ to be (u, v)'s cost to LP 2.2 in the objective, i.e., $x_{uv} = \mathsf{LP}(u, v)$ for $(u, v) \in E^+$ and $x_{uv} = 1 - \mathsf{LP}(u, v)$ if $(u, v) \in E^-$. Similarly, we define a cost for each edge with respect to Algorithm 1, $\mathsf{ALG}(u, v) = \mathbb{1}((u, v)$ is a disagreement). Then summing over all edges incident to vertex u, we obtain the disagreements to each vertex, $\mathsf{ALG}(u) = \sum_{v \in V} \mathsf{ALG}(u, v)$.

The technical work of Kalhan, Makarychev, and Zhou [2019] is in showing that

$$\mathsf{ALG}(u) = \sum_{v \in V} \mathsf{ALG}(u, v) \leqslant 5 \cdot \sum_{v \in V} \mathsf{LP}(u, v) = 5y(u). \tag{6}$$

Davies, Moseley, and Newman [2023] discuss the run-time of Algorithm 1 (see Appendix A in their paper). The rounding algorithm has run-time $O(n^2)$ for general graphs, and the full run-time of the KMZ algorithm is dominated by the time to solve the LP.